

FML Unit-3 Syllabus -

1) Clustering:

- Centroid based Clustering ✓
- Density based Clustering ✓
- Distribution based Clustering and Hierarchical Clustering ✓
- K Means Clustering ✓
- KNN ✓
- DBSCAN clustering algorithm ✓
- Performance metrics for clustering: Silhouette Score ✓

2) Association Rule Learning:

- Apriori algorithm ✓
- FP Growth Algorithm ✓
- Applications of Association Rule Learning ✓
- Market Basket Analysis

3) Reinforcement Learning: ✓

- Types of Reinforcement Learning ✓
- Key Features of Reinforcement Learning ✓
- Elements of Reinforcement Learning ✓
- Applications of Reinforcement Learning ✓

Unit - 3

Data Clustering -

- 1) Unsupervised Machine Learning
- 2) Designed to group similar data points together
- 3) Applications:
 - customer segmentation to identify ~~various~~ distinct customer groups
 - anomaly detection for identifying unusual patterns
 - document classification for organizing for large text corpora

Centroid Based Clustering -

- 1) Partitions data into a predetermined number of 'K' clusters with each cluster being defined by its central point, or centroid
 - 2) Primary objective is to minimize the sum of squared distances between individual data points & the centroid of their assigned cluster, aiming for compact & well separated clusters
 - 3) Advantages - ^{efficient}
 - (i) Computationally ~~expensive~~
 - (ii) Simple to implement
- | | |
|---|--|
| Disadvantages -
(i) Requires predefining the no. of clusters (K)
(ii) Sensitive to initial centroid placement
(iii) Assumes clusters are spherical or convex | |
|---|--|

Density Based Clustering -

- 1) Divides data into a fixed number of K clusters
- 2) Each cluster is represented by a centroid (central point)
- 3) Minimizes the sum of squared distance between data points & their cluster centroid
- 4) Aims to form compact & well separated clusters
- 5) Advantages -
 - (i) ~~Highly~~ computationally efficient
 - (ii) ~~Sensitive to~~ Simple &

Disadvantages

Density Based Clustering -

- 1) Identifies clusters as regions of high density data points
- 2) Clusters are separated by areas of low density
- 3) Effective for finding complex or irregular cluster shapes
- 4) Cluster grows from core points (points having enough neighbors within a certain radius)
- 5) Points not in any dense region are treated as noise

6) Advantages

(i) Can discover arbitrary shaped clusters

(ii) Robust to noise & can detect outliers

(iii) No need to predefine no. of clusters (K)

Disadvantages

(i) Struggles with varying densities across clusters

(ii) Performance may drop in high dimensional data

(iii) Sensitive to parameters ϵ and MinPts

K-Means: Workhorse of Centroid Based Methods -

1) Algorithm:

Assigns data points to nearest centroid, updates centroids as mean of points, & repeats until stable

2) Performance Metric:

Uses Sum of Squared Errors (SSE) or inertia - lower value means better clustering

3) Use Case:

Commonly used to segment e-commerce customers for targeted marketing.

4) Convergence & Optimal K :

Usually stabilizes in 5-8 iterations. Optimal K is found via Elbow or Silhouette Method

DBSCAN: Discovering Arbitrary Shapes -

1) Algorithm:

Groups points as core, border, or noise using 2 parameters - epsilon (ϵ) and MinPts

2) Parameter Tuning:

Proper ϵ and MinPts depend on data types and dimensions, tuning often needs domain knowledge

3) Use Case:

Detects fraud patterns or anomalies in large datasets

4) Effectiveness:

Finds non-spherical clusters (crescent/ring shaped) and handles noise/outliers effectively.

Centroid vs Density Based.

Aspect	Centroid Based	Density Based
1) Cluster Shape	Spherical or convex clusters	Arbitrary shapes ("S", "C", patterns)
2) Noise Handling	Assigns all points to a cluster; sensitive to outliers	Explicitly identifies outliers as 'noise' points
3) Parameter Sensitivity	Requires predefining k	Needs ' ϵ ' (neighborhood radius) and MinPts, often more challenging to tune
4) Scalability	Generally faster for very large datasets [$O(n \cdot k \cdot i)$]	Can be slower due to neighborhood queries $O(n \log n)$ or $O(n^2)$
5) Use Cases	Customer segmentation, image compression	Spatial data analysis, anomaly detection, fraud detection

Distribution Based Clustering-

- 1) Models data using probability distributions
- 2) Assigns points to clusters based on probability of belonging
- 3) Can model complex cluster shapes
- 4) Provides soft clustering (partial membership)
- 5) Works well ~~with~~ when data follows Gaussian like patterns
- 6) Handles overlapping clusters better than centroid methods
- 7) Limitation: Computationally expensive & sensitive to initialization
- 8) Advantages:
 - (i) Models complex cluster shapes using probability distributions
 - (ii) Supports soft clustering
 - (iii) Works well for statistically distributed data (eg. Gaussian)
 - (iv) Handles overlapping clusters
- 9) Disadvantages:
 - (i) Assumes data fits a specific distribution
 - (ii) Sensitive to initialization
 - (iii) Computationally expensive
 - (iv) Struggles with high dimensional data without reduction
- 10) Applications:

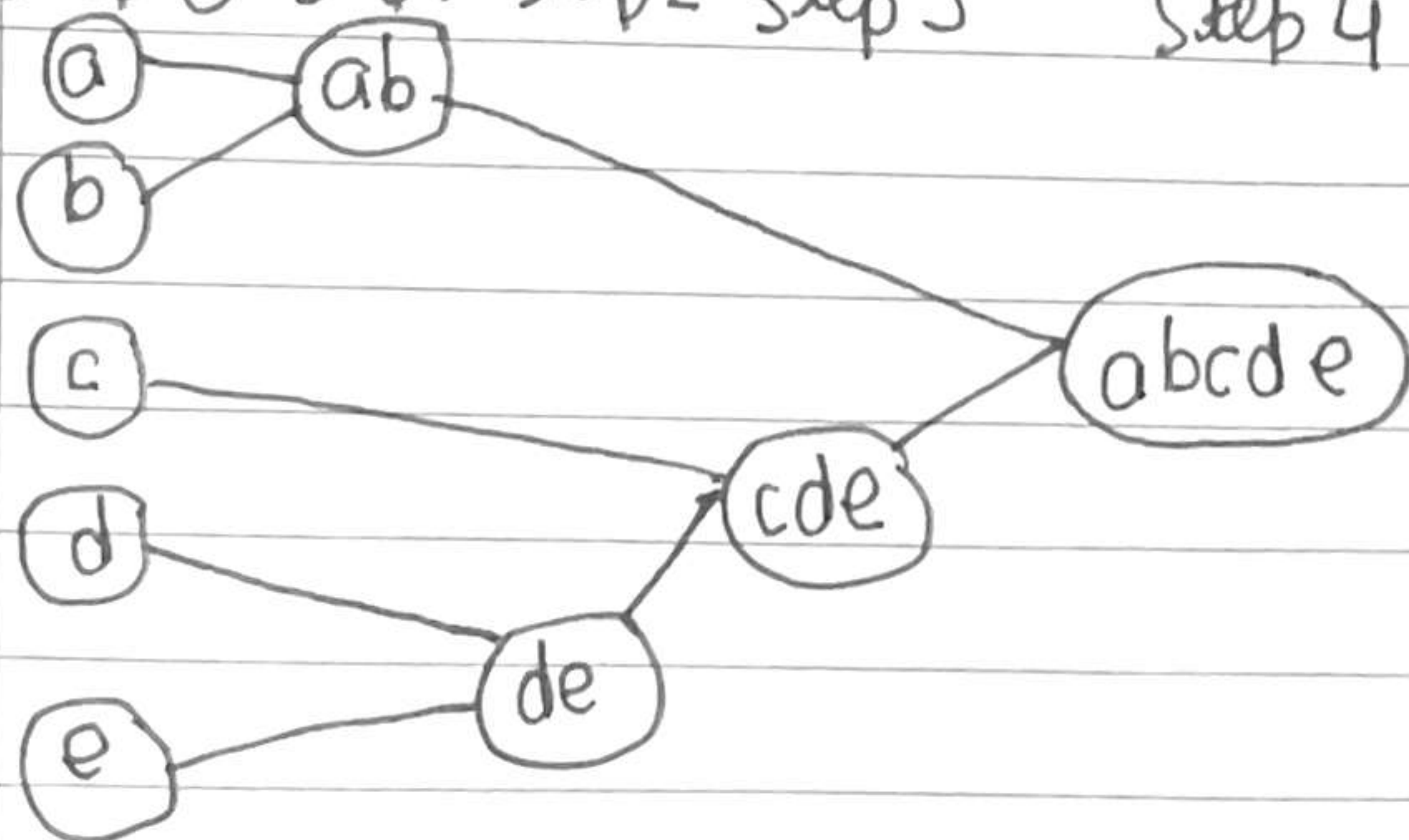
(i) Image segmentation (ii) Speech recognition (iii) Bioinformatics (iv) Market segmentation	(v) Anomaly detection (vi) Document classification
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Hierarchical Clustering-

- 1) Builds a tree like hierarchy of clusters
- 2) 2 types
 - (i) Agglomerative (bottom up)
 - (ii) Divisive (top down)
- 3) No need to predefine no. of clusters

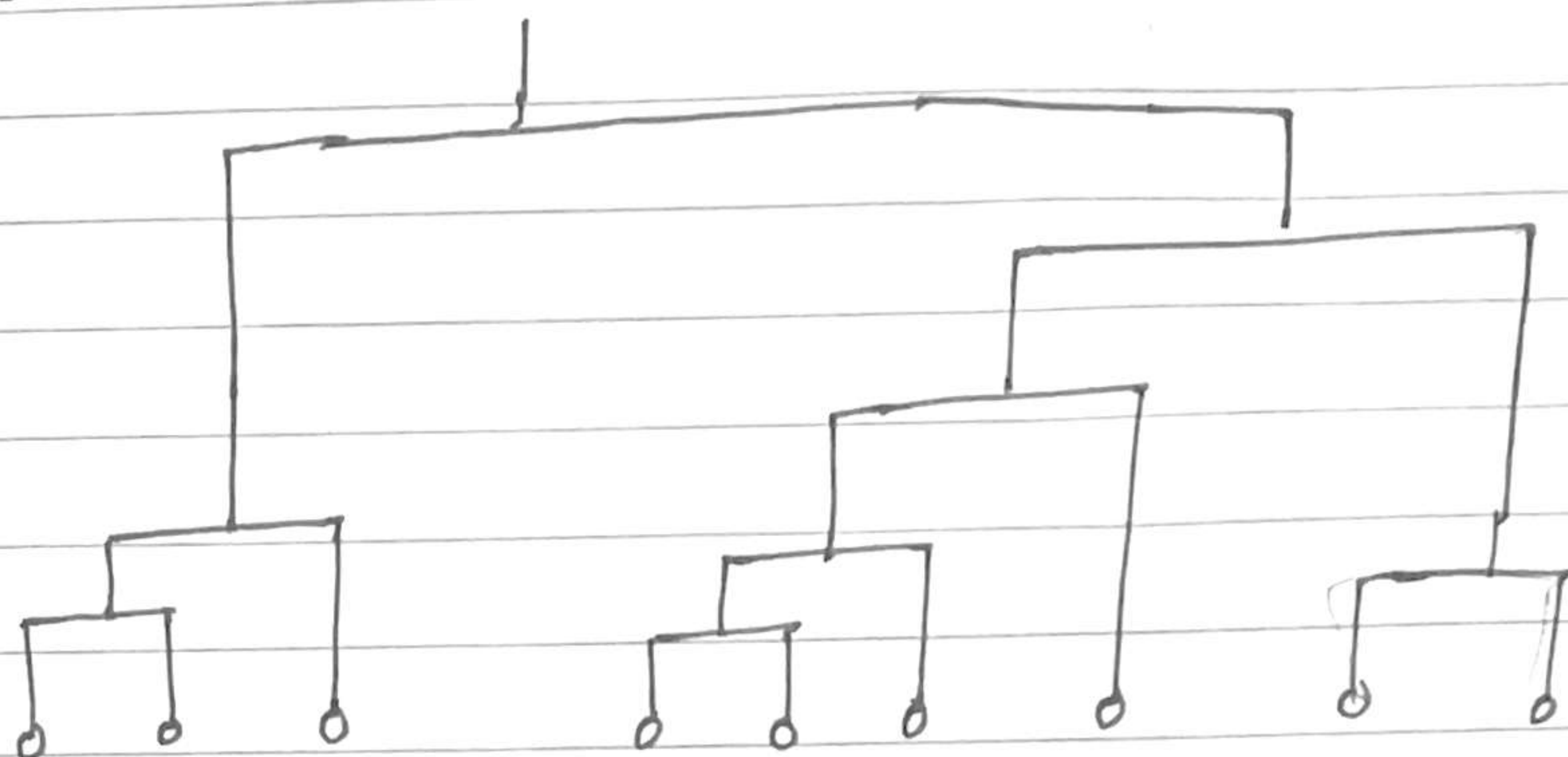
- 4) Produces a dendrogram for visualization
- 5) Easy to implement & interpret
- 6) Limitation: Computationally expensive for large data
- 7) Sensitive to noise, outliers, and linkage method
- 8) Advantages:
 - (i) Easy to understand & implement
 - (ii) No need to predefine cluster count
 - (iii) Give dendrogram for visualization
 - (iv) Works with different shapes & small datasets
- 9) Disadvantages:
 - (i) Slow & costly for large datasets
 - (ii) Sensitive to noise & outliers
 - (iii) No backtracking once merged/split
 - (iv) Results depend on linkage method
 - (v) Not good for high dimensional data
- 10) Use distance matrix as clustering criteria. This method does not require the number of clusters K as an input, but needs a termination condition

Step 0 Step 1 Step 2 Step 3 Step 4 Agglomerative (AGNES)



Step 4 Step 3 Step 2 Step 1 Step 0 Divisive (DIANA)

- 11) Decompose data objects into a several levels of nested partitioning (Tree of clusters), called a dendrogram
- 12) A clustering is obtained by cutting the dendrogram at the desired level, then each connected component forms a cluster.



K Means Clustering -

- 1) Unsupervised ML algorithm designed to automatically group similar data points together
- 2) Divides data into K clusters based on similarity
- 3) Goal: Minimize the within cluster sum of squares (WCSS)
- 4) Each point is assigned to the nearest cluster centroid
- 5) Assumes clusters are spherical & equal in size
- 6) Works best when these assumptions hold true
- 7) K Means algorithm:
 - (i) Choose K (no. of clusters)
 - (ii) Initialize centroids randomly
 - (iii) Assign data points to nearest centroid (using Euclidean distance)
 - (iv) Update centroids - recalculate mean of each cluster
 - (v) Reassign points - update clusters based on new centroids
 - (vi) Converge - repeat until centroids stabilize or max iterations reached

Choosing k and evaluating cluster quality -

- 1) Elbow method: Plot WCSS vs k ; elbow point gives optimal k
- 2) Silhouette Score: measures how well points fit within their cluster (range -1 to 1)
- 3) Inertia: Sum of squared distances; lower = better clustering
- 4) Choose k using domain knowledge or business goals

Practical applications of k -Means -

- | | |
|--------------------------|------------------------|
| 1) Customer Segmentation | 4) Anomaly detection |
| 2) Document clustering | 5) Geospatial analysis |
| 3) Image compression | |

Advantages -

- 1) Simplicity & interpretability
- 2) Computational efficiency
- 3) Clear Cluster Boundaries

Limitations -

- 1) Sensitivity to initial centroids
- 2) Spherical cluster assumption
- 3) Requires pre specified (k)
- 4) Outlier sensitivity
- 5) Varying density / size

KNN

KNN Algorithm Steps

- 1) Choose $K \rightarrow$ no. of nearest neighbors
- 2) Calculate distance between query point & all training data points
- 3) Find the K nearest neighbors based on smallest distances
- 4) Classification $\rightarrow$ assign class with majority vote
- 5) Regression $\rightarrow$ takes average of neighbor's values

Parameter selection & preprocessing -

1) Choosing k :

Small $k \rightarrow$ overfitting, noisy boundaries

Large $k \rightarrow$ smoother boundaries, risk of underfitting

2) Distance Metric: affects clustering of neighbors

3) Preprocessing needs:

Normalization / Standardization $\rightarrow$ essential since KNN is distance based

4) Handle missing values, remove irrelevant features

Advantages & Limitations of KNN -

1) Advantages:

(i) Easy to understand & implement

(ii) Non parametric $\rightarrow$ no assumption on data distribution

(iii) Works for multi class problems

2) Limitations:

(i) High computational cost

(ii) Requires feature scaling

(iii) Not suitable for high dimensional data

Applications of KNN -

1) Image recognition, handwriting detection

2) Recommendation systems

3) Medical diagnosis

DBSCAN -

1) Core Point: Has at least MinPts points within

ϵ (epsilon) radius - forms cluster core

2) Border Point: Lies within ϵ of a core point but has fewer than MinPts - extends clusters

- 3) Noise Point: Neither core nor border - considered an outlier
- 4) Main idea: clusters are ~~form~~ formed based on density connectivity, identifying arbitrary shapes & isolating noises

Key Parameters:

1) Epsilon (ϵ):

Defines radius around a point to find neighbors
- controls cluster density

2) MinPts:

Min no. of points required within ϵ to form a core point
(usually 4-6)

* Proper selection of ϵ and MinPts is crucial;
often found using the K distance graph

DBSCAN algorithm (Steps) -

- 1) Select point: Pick an unvisited data point
- 2) Find neighbors: Identify all points within ϵ radius
- 3) Core check: If neighbors $\geq$ MinPts $\rightarrow$ core point
- 4) Expand cluster: add all reachable points to the cluster
- 5) Border Point: add & but don't expand further
- 6) Noise: If not core or border $\rightarrow$ mark as noise
- 7) Repeat: Continue until all points are visited

Arbitrary shapes & noise -

- 1) Detects clusters of any shape
- 2) Unlike K means, it handles non spherical & intertwined clusters
- 3) Outliers are identified as noise instead of forcing them into clusters
- 4) Works well for real world data with irregular patterns & anomalies

Advantages -

- 1) Handles arbitrary shapes
- 2) No pre defined K
- 3) Robust to Noise

Limitations -

- 1) Parameters sensitivity
- 2) High dimensionality
- 3) Computational cost
- 4) Difficult with varying densities

Applications -

- 1) Geographic Information System (GIS)
- 2) Anomaly Detection
- 3) Customer segmentation
- 4) Traffic data analysis
- 5) Image processing
- 6) Healthcare & Bioinformatics

Cluster Validity -

- 1) Every clustering algorithm produces a set of clusters
- 2) Evaluating their 'goodness' is challenging - clusters depend on interpretation
- 3) Why evaluate?
 - (i) To evaluate different clustering algorithms
 - (ii) To compare 2 clustering results on same data
 - (iii) To compare individual clusters
 - (iv) To check for noise or outliers in the data

Examples (On Numericals) -

1) K-Means Clustering -

$$WCSS = \sum \sum ||x - \mu_i||^2$$

WCSS = Within Cluster Sum of Squares
 μ_i = centroid of cluster

Given points on 1D axis: [1, 2, 5, 8] and $K=2$

Initial centroids: 1 and 5

Step 1.

Point	Distance to 1	Distance to 5	Assigned Cluster
1	0	4	Cluster 1
2	1	3	Cluster 1
5	4	0	Cluster 2
8	7	3	Cluster 2

Clusters formed:

Cluster 1 = [1, 2]

Cluster 2 = [5, 8]

Step 2. Recompute centroids

$$C_1 = (1+2)/2 = 1.5$$

$$C_2 = (5+8)/2 = 6.5$$

New centroids $\rightarrow$ 1.5 & 6.5

Step 3. Reassign points

Point	Distance to 1.5	Distance to 6.5	Assigned Cluster
1	0.5	5.5	Cluster 1
2	0.5	4.5	Cluster 1
5	3.5	1.5	Cluster 2
8	6.5	1.5	Cluster 2

$\Rightarrow$ Clusters remain same, convergence achieved

Manhattan Distance (K-Medoids):

2) Data points: $(1,3)$, $(2,2)$, $(3,4)$, $(5,6)$, $(8,8)$

$$\begin{array}{c} \uparrow \\ \text{Medoid} \\ M1 \end{array}$$

$$\begin{array}{c} \uparrow \\ \text{Medoid} \\ M2 \end{array}$$

$A = (1,3)$

$B = (2,2) = M1$

$C = (3,4)$

$D = (5,6) = M2$

$E = (8,8)$

For point $A(1,3)$:

$d(A, M1) = |1-2| + |3-2| = 1+1 = 2$

$d(A, M2) = 4+3 = 7$

Assign to Cluster 1

For point $B(2,2)$:

$d(B, M1) = 0$

$d(B, M2) = 7$

Assign to cluster 1

For point $C(3,4)$

$d(C, M1) = 1+2 = 3$

$d(C, M2) = 2+2 = 4$

Assign to Cluster 1

For point $D(5,6)$:

$d(D, M1) = 1+2 = 3$

$d(D, M2) = 2+2 = 4$

Assign to cluster 2

For point E

$d(E, M1) = 6+6 = 12$

$d(E, M2) = 3+2 = 5$

Assign to Cluster 2

Initial clusters:

Cluster 1: $\{A(1,3), B(2,2), C(3,4)\}$ Cluster 2: $\{D(5,6), E(8,8)\}$

Initial Total Cost:

$$\text{Cost} = d(A, M1) + d(B, M1) + d(C, M1) + d(D, M2) + d(E, M2)$$

$$= 2 + 0 + 3 + 0 + 5$$

$$= 10$$

Current cost of cluster 1

$$: d(A,B) + d(B,B) + d(C,B) = 2 + 0 + 3 = 5$$

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Now, swap medoids

If A is medoid:

$$\text{Cost} = d(A,A) + d(B,A) + d(C,A)$$

$$d(B,A) = |2-1| + |2-3| = 2$$

$$d(C,A) = 2 + 1 = 3$$

$$\text{Total cost} = 0 + 2 + 3 = 5$$

If C is medoid

$$\text{Cost} = d(A,C) + d(B,C) + d(C,C)$$

$$d(A,C) = 2 + 1 = 3$$

$$d(B,C) = 1 + 2 = 3$$

$$\text{Total cost} = 3 + 3 + 0 = 6$$

$$\text{Cost for (Medoid B)} = 5$$

$$\text{" " (Medoid A)} = 5$$

$$\text{" " (Medoid C)} = 6$$

Min = 5 $\Rightarrow$ Keep Medoid as B

Current cost of cluster 2

$$: d(D,D) + d(E,D) = 0 + 5 = 5$$

If E is medoid:

$$\text{Cost} = d(D,E) + d(E,E)$$

$$d(D,E) = 3 + 2 = 5$$

$$\text{Total cost} = 5 + 0 = 5$$

$$\text{Cost for (Medoid D)} = 5$$

$$\text{Cost for (Medoid E)} = 5$$

$$\text{Min} = 5 \Rightarrow \text{Keep Medoid as D}$$

Algorithm has converged

$$\text{Total cost} = 5 + 5 = 10$$

DBSCAN:

3) Data Points: $P_1(1,1)$, $P_2(2,2)$, $P_3(2,3)$, $P_4(8,8)$, $P_5(8,9)$, $P_6(25,80)$

Parameters: $\epsilon = 2$ (max radius of neighborhood)

MinPts = 2 (min points in neighborhood)

Distance : we use Euclidean metric

Core Point: A point that has atleast MinPts (2) neighbors within its ϵ neighborhood (radius 2). The point itself is included in the neighbor count

Noise Point: A point is neither a core point nor a border point

Border Point: A point that is not a core point but is in the ϵ neighborhood of a core point

Step 1. Find neighbors for each point

(i) Point $P_1(1,1)$:

$$d(P_1, P_1) = 0 \leq 2$$

$$d(P_1, P_2) = \sqrt{1+1} = \sqrt{2} = 1.414 \leq 2$$

$$d(P_1, P_3) = \sqrt{1+4} = \sqrt{5} = 2.236 > 2$$

(P_4, P_5, P_6 are all much further than 2)

Neighbors: $\{P_1, P_2\}$

Count: 2

Since $2 \geq \text{MinPts}(2)$, P_1 is a core point

(ii) Point $P_2(2,2)$:

$$d(P_2, P_1) = \sqrt{2} = 1.414 \leq 2$$

$$d(P_2, P_2) = 0 \leq 2$$

$$d(P_2, P_3) = \sqrt{0+1} = 1 \leq 2$$

(P_4, P_5, P_6 are all much further than 2)

Neighbors: $\{P_1, P_2, P_3\}$

Count: 3

Since $3 \geq \text{MinPts}(2)$, P_2 is a core point

(iii) Point $P_3(2,3)$:

$$d(P_3, P_1) = \sqrt{5} = 2.236 > 2$$

$$d(P_3, P_2) = 1 \leq 2$$

$$d(P_3, P_3) = 0 \leq 2$$

(P_4, P_5, P_6 are all much further than 2)

Neighbors: $\{P_2, P_3\}$

Count: 2 $\Rightarrow P_3$ is a core point

(iv) Point $P_4(8,8)$:

P_1, P_2, P_3 are all much further than 2)

$$d(P_4, P_4) = 0 \leq 2$$

$$d(P_4, P_5) = \sqrt{0+1} = 1 \leq 2$$

$$d(P_4, P_6) = \sqrt{289+5184} > 2$$

Neighbors: $\{P_4, P_5\}$

Count: 2 $\Rightarrow P_4$ is a core point

(v) Similarly P_5 is also a core point

(vi) Point $P_6(25,80)$

(All other points are very far away)

$$d(P_6, P_6) = 0 \leq 2$$

Neighbors: $\{P_6\}$

Count: 1

Since $1 < \text{MinPts}(2)$, P_6 is not a core point

Now, to be a border point $\Rightarrow P_6$ must be in the
 $\epsilon \in$ neighborhood (radius 2)
 of a core point ($P_1, P_2, P_3,$
 P_4 or P_5)

But, no distance were ≤ 2
 (except for itself)

$\Rightarrow P_6$ is a noise point

Final Results:

Core Point: $\{P_1, P_2, P_3, P_4, P_5\}$

Border Points: None

Noise Points: $\{P_6\}$